

Stylometrics of Fake News

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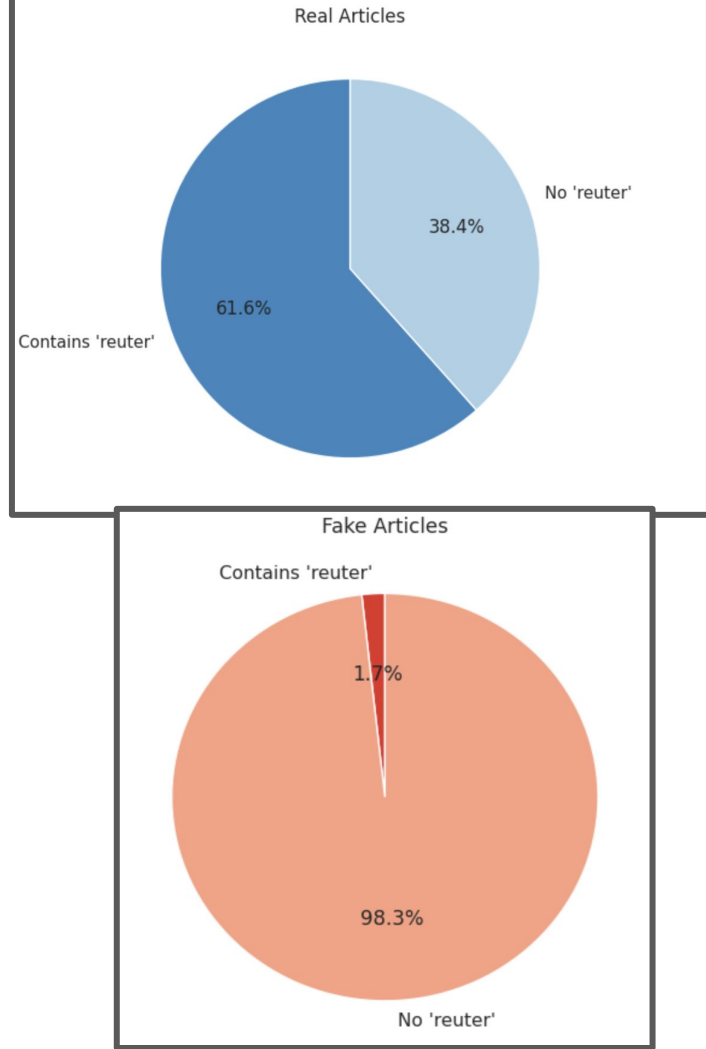
Stylometry: study of linguistic style, usually to written language.

The Problem

- “Fake vs. Real” can easily collapse to “Which outlet published this”
- Models exploit shortcuts: outlet names, recurring entities, etc.
- Can lead to high accuracy on the training data, but poor generalization
- Public Datasets leak information allowing models to achieve high accuracy classification results solely on the origin of content (i.e. site/author) and not on the actual content.

Leakage and Misleading results

- 95% accuracy with TF-IDF (Default Params)
- Model exploited the source of an article.
- Real articles came from Reuters; fakes didn't.
- **Shortcut Learning**: when a model doesn't solve the intended task but can achieve high scores



WelFake Dataset

Word Embedding over Linguistic Features for Fake News Detection

real	fake
35,028	37,106

- 72,134 news articles
- Merged from Kaggle, McIntire, Reuters, BuzzFeed Political

Paper: WELFake: Word Embedding Over Linguistic Features for Fake News Detection. DOI 10.1109/TCSS.2021.3068519

- WELFake model ~ 96.73%
- 62,348 unique values; Columns: Title, Text, Label; [WelFake](#)

TF-IDF vs Engineered features

Can hand-crafted *style* features capture signal that generalizes better than vocabulary?

- TF-IDF Measure of importance of a word to a document measure

$$\text{idf}(t, D) = \log \frac{N}{n_t}$$

- IDF: Inverse Document Frequency

- The specificity of a term can be quantified as an inverse function of the number of documents in which it occurs.

- TF: Term Frequency

- The relative frequency of a term within a document

$$\text{tf}(t, d) = \frac{f_{t,d}}{\sum_{t' \in d} f_{t',d}}$$

- We Engineered Features to capture writing style and structure of an article.
- For example: exclamation rate is the rate '!' appears over the total number of words.

Experiment Setup

- Same Train/test split used for each model
 - Real: 45%
 - Fake: 55%
- evaluation metric: AUC
- Model Params Constant
- Same Metric Harness used for each model
- Cross-validated scores with standard deviation.

Three XGBoost Models

1. TF-IDF (Baseline)
2. Stylometric (Features Engineered)
3. Combined (Stylometric + TF-IDF)

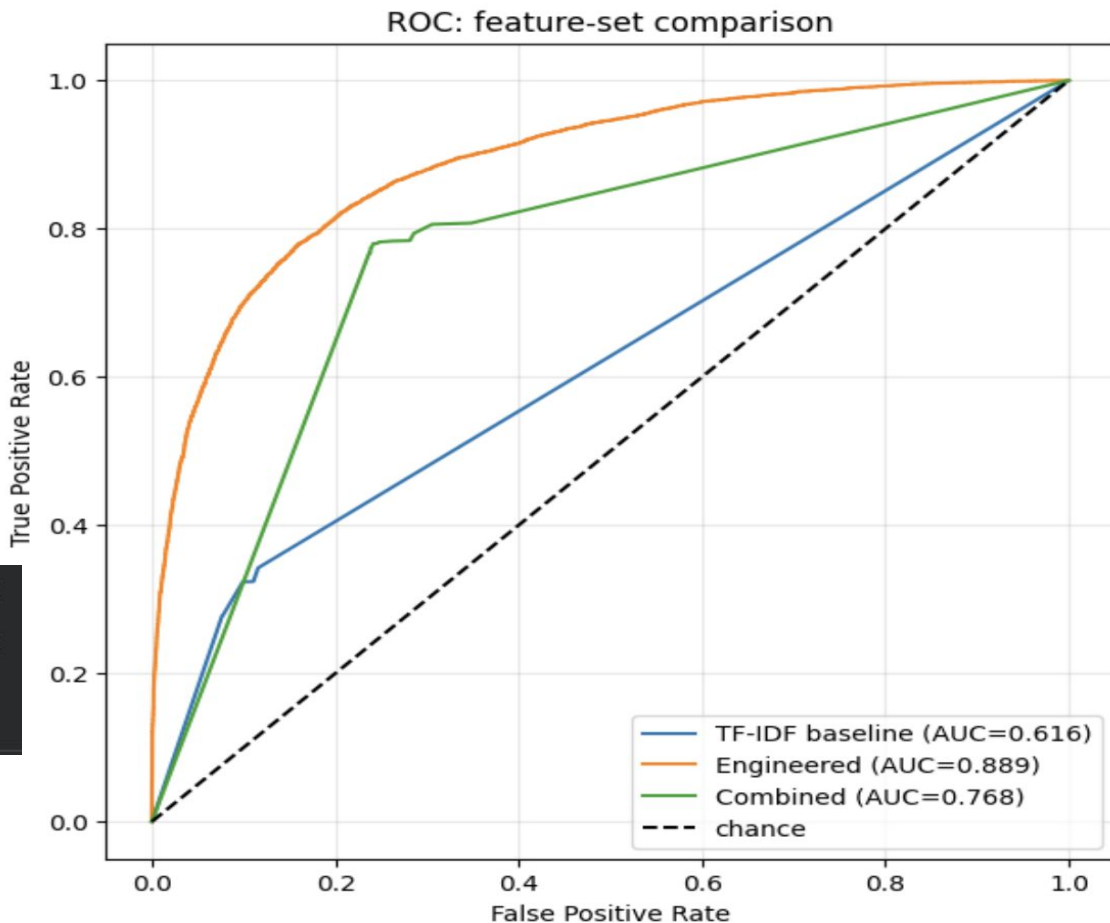
Results

Engineered Features outperform
TF-IDF baseline

Combination of Features with
TF-IDF produces a noisy feature
space, degrading performance

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CROSS-VALIDATED AUC (mean +/- std over folds)
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TF-IDF (cleaned)	CV AUC: 0.6544 +/- 0.0543
Engineered	CV AUC: 0.8904 +/- 0.0006
TF-IDF + engineered	CV AUC: 0.6676 +/- 0.0898



What the Features Reveal

- Features were designed modularly so that documents with different available fields could still be processed consistently by the model.
- This makes the system more robust for practical use.
- The main focus of features were basically the language and patterns used in the text.

These features and their AUC results reveal that using the raw text format may be a strong indicator of discovering whether an article is real or fake.

n_words	word count
avg_word_len	average word length
exclaim_rate	ratio of exclamation marks
Question_rate	ratio of question marks
type_token_ratio	vocabulary diversity
pronoun_rate	ratio of pronouns
certainty_rate	ratio of certainty words
cedging_rate	ratio of hedging words
superlative_rate	ratio of superlative words
sensational_rate	ratio of sensational words

Engineered features results

feature_auc_results (1)

1	Features Used	AUC
2	n_words + n_sentences + avg_word_len + avg_sent_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + digit_rate + type_token_ratio	0.8719008757745040
3	n_words + n_sentences + avg_word_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + digit_rate + type_token_ratio	0.8710890009015860
4	n_sentences + avg_word_len + avg_sent_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + digit_rate + type_token_ratio	0.870380668298138
5	n_words + avg_word_len + avg_sent_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + digit_rate + type_token_ratio	0.8702286492413550
6	n_words + avg_word_len + avg_sent_len + exclaim_rate + question_rate + caps_word_rate + digit_rate + type_token_ratio	0.8661124509063330
7	n_words + n_sentences + avg_word_len + avg_sent_len + exclaim_rate + question_rate + caps_word_rate + digit_rate + type_token_ratio	0.8661004965677810
8	n_words + n_sentences + avg_word_len + exclaim_rate + question_rate + caps_word_rate + digit_rate + type_token_ratio	0.8644530748644140
9	n_words + n_sentences + avg_word_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + digit_rate	0.8639145378306170
10	n_words + avg_word_len + avg_sent_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + digit_rate	0.8637881464859510
11	n_words + n_sentences + avg_word_len + avg_sent_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + digit_rate	0.8631879440027900
12	n_sentences + avg_word_len + avg_sent_len + exclaim_rate + question_rate + caps_word_rate + digit_rate + type_token_ratio	0.8622636555125080
13	n_words + n_sentences + avg_word_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + type_token_ratio	0.8609688727059870
14	n_words + n_sentences + avg_word_len + avg_sent_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + type_token_ratio	0.8605846092011190
15	n_sentences + avg_word_len + avg_sent_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + digit_rate	0.860557223000729
16	n_sentences + avg_word_len + avg_sent_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + type_token_ratio	0.8601569696521670
17	n_words + avg_word_len + avg_sent_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + type_token_ratio	0.8591247652164530
18	n_words + n_sentences + avg_word_len + avg_sent_len + exclaim_rate + question_rate + caps_word_rate + digit_rate	0.8569054459277140
19	n_sentences + avg_word_len + avg_sent_len + exclaim_rate + question_rate + caps_word_rate + digit_rate	0.8565592491850250
20	n_words + n_sentences + avg_sent_len + exclaim_rate + question_rate + quote_rate + caps_word_rate + digit_rate + type_token_ratio	0.8562893099626640
21	n_words + avg_word_len + avg_sent_len + exclaim_rate + question_rate + caps_word_rate + type_token_ratio	0.8561891888546740
22	n_words + avg_word_len + avg_sent_len + exclaim_rate + question_rate + caps_word_rate + digit_rate	0.8558864019731500

Future work

- Expand the engineered feature set to include additional writing-style indicators.
- Evaluate larger combinations of engineered features using cross-validated AUC.
- Analyze why certain feature combinations achieve significantly higher performance.
- Compare additional engineered-feature performance against TF-IDF and combined feature approaches.
- Test whether the discovered feature combinations remain effective on unseen news sources.